

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: CY

2. Subject Name: Structure & function of Biomolecules

L-T-P: 3-1-0 Credits: 4

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus:

Amino acid, peptide and proteins: (10L)

Configuration, classification and nomenclature of amino acids; acid-base properties of amino acids; isoelectric point; peptide bonds and disulfide bonds; chemical synthesis of peptide via N-protection and C-activation; automated peptide synthesis; introduction to protein structures; determination of the primary structure (sequence) of proteins (including application of MS); dihedral angles and secondary structure of proteins: Ramachandran plot (briefly); regular structures (α -helix, β -sheet; β -turn); tertiary and quaternary structure of proteins

Coenzymes and introduction to enzymes: (8L)

Structures and function of coenzymes: NAD⁺/NADH; FAD/FM; thiamine pyrophosphate; biotin; PLP: coenzyme B₁₂; THF; vitamin K₂; enzymes as biocatalysts of cells; classification of enzymes, active site; Michaelis-Menten kinetics, drugs as inhibitor of enzymes.

Carbohydrates: (8L)

The D- and L- notations; aldoses; ketoses; cyclic hemiacetals; mutarotation; formation of glycosides; anomeric effect; reducing and nonreducing sugars; redox reactions of monosaccharides; lengthening the chain: Kiliani-Fischer synthesis; shortening the chain: Wohl degradation; disaccharides and polysaccharides including chitin and peptidoglycans; structure and function of sialic acid; carbohydrates on cell surfaces.

Nucleosides, nucleotides, and nucleic acids: (8L)

Nucleosides and nucleotides as subunits of nucleic acids; stability of DNA vs RNA; why DNA contains thymine instead of uracil; chemical synthesis of DNA (synthesizer); DNA sequence determination; WC base pairing; DNA- double helix; three kinds of RNA; Central Dogma; replication; transcription; translation; PCR; genetic engineering(time permitting).

Chemistry of metabolism: (8L)

ATP and chemical energy; pseudorotation of phosphorous; mechanisms for phosphoryl transfer reactions; "high-energy" of phosphoanhydride bonds; stability of ATP in a cell; introduction to metabolism; four stages of catabolism; catabolism of fats; catabolism of carbohydrates; fates of pyruvate; catabolism of proteins; citric acid cycle; oxidative phosphorylation; biosynthesis of L-tyrosine, L-phenylalanine, L-histidine; purine and

pyrimidine bases.

Cells as compartments: (6L)

Lipids; fatty acids and their types, structure and biological functions of various class of lipids – triacylglycerol, phospholipids, glycolipids, sphingolipids, terpenoid lipids, (including steroids); biological membranes.

Reference Books:

1. Organic Chemistry by Paula Bruice, Pearson publisher, 8th edition, 2016
2. Bioorganic Chemistry by Herman Dugas, Springer India, 3rd edition, 2005
3. Nucleic Acids in Chemistry and Biology by G Michael Blackburn, Michael J Gait, David Loakes, David M Williams, RSC publishing, 3rd edition, 2006
4. Biochemistry by Donald Voet and Judith Voet, Wiley, 4th Edition, 2010

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Amino acid, peptide and proteins	10
2.	Coenzymes and introduction to enzymes	8
3.	Carbohydrates	8
4.	Nucleosides, nucleotides, and nucleic acids	8
5.	Chemistry of metabolism	8
6.	Cells as compartments	6
Total number of hours		48